

Micro I, class 15

An expected utility representation is a function $U : \mathcal{L} \rightarrow [0, 1]$ that represents $\succsim \subset \mathcal{L}^2$ and takes the form

$$U(p) = \sum_i U(e_i) p_i, \quad (1)$$

or, setting $u(x_i) = U(e_i)$, $U(p) = \sum_i u(x_i) p_i$, as in my Theorem statement.

We proved a simplified version of the expected utility theorem (with redundant Axioms A4) and A5) simply assumed to be true). The representation $U : \mathcal{L} \rightarrow [0, 1]$ is defined implicitly by

$$p \sim U(p)e_n + (1 - U(p))e_1 \quad (2)$$

where e_n is a best and e_1 is a worst element of $\mathcal{L}$, the set of distributions on the outcome set X . The crucial property that the independence axiom implies is that it is *affine*, that is, both convex and concave:

$$U(\lambda q + (1 - \lambda)p) = \lambda U(q) + (1 - \lambda)U(p) \quad \text{for all } q, p \in \mathcal{L} \text{ and } \lambda \in [0, 1], \quad (3)$$

which leads directly to the expected-utility form, $U(p) = \sum_i U(e_i) p_i$, by induction. The function U is called a *von Neumann-Morgenstern (vN-M) utility*. I will follow a common abuse of terminology and refer to the function u on X defined by $u(x_i) = U(e_i)$ as a vN-M utility.

That U is *affine* (both convex and concave) implies that the *indifference sets* themselves are convex, in particular indifference curves for a three-outcome set are *straight lines* with *common slope*.

We used this geometric property to show that *modal* choices in the *Allais paradox* are not consistent with expected-utility preferences. The outcome set is $X = \{w, w + 1, w + 5\}$, where the units are millions of dollars. Consider $p = (0, 1, 0)$ vs $q = (.01, .89, .1)$. And $r = (.89, .11, 0)$ vs $s = (.9, 0, .1)$. In “experiments” (often just surveys) the *modal* response is $p \succ q$ and $s \succ r$. We verified graphically that these choices cannot be explained by expected-utility preferences.¹ I will ask you to show *directly* that the modal choice is inconsistent with the *independence axiom*.

Next up, I will specialize to monetary outcomes, but will replace the finite outcome set $X = \{x_1, \dots, x_n\}$ with the infinite outcome set $X = \mathbb{R}_+$. I will take up three questions: what is a good definition of *risk aversion*; what is a good definition of one person being *more risk averse* than another person; and what is a good definition of an *increase in risk*? I will illustrate with examples from portfolio choice. (You will pursue an example from insurance demand.)

ADDENDUM I: TWO ALTERNATIVES TO EXPECTED UTILITY

Two alternative theories of decision making under uncertainty are *mean-variance preferences* (MV) and *cumulative prospect theory* (CPT). Each can accommodate the modal preferences in the Allais example.

Mean-variance representations usually specialize to the case of monetary outcomes and are functions just of the mean (μ) and standard deviation (σ) of a risk. A *risk neutral* person cares only about the mean wealth, and so is consistent with both MV and EU preferences. The function $(\star) \tilde{U}(\mu, \sigma) = \mu - \lambda\sigma$ for $\lambda > 0$ is a simple example of a MV representation that is not consistent with EU.²

¹Both MWG and Kreps present and discuss the Allais paradox, as does Mark Machina in the *JEL* piece I posted to Canvas.

²To illustrate the inconsistency, consider the two asset portfolio problem with safe rate of return normalized to 0. For EU, demand for the risky asset is positive if the expected rate of return of the risky asset is positive, as we confirmed together. The implication is false for the function $(\star)$: letting μ_x be the expected rate of return on the risky asset and σ_x its standard deviation, if $\lambda\sigma_x > \mu_x > 0$, then the demand for the risky asset is 0 even though the expected rate of return on the risky asset is positive. My paper with Mike Ormiston, “Mean-variance preferences and investor behaviour,” *The Economic Journal* 111.474 (2001): 849-861, points out some surprising differences between mean-variance and expected-utility preferences.

Cumulative prospect theory (CPT) combines elements of Kahneman and Tversky’s *prospect theory*—Kahneman, Daniel, and Amos Tversky. “Prospect Theory: An Analysis of Choices Involving Risk,” *Econometrica* 47 (1979): 263-291—and Quiggin’s *rank-dependent expected utility* (RDEU)—Quiggin, J. (1982), ‘A theory of anticipated utility’, *Journal of Economic Behavior and Organization* 3(4), 323–43. It asserts that preferences over distributions of monetary outcomes are represented by

$$U(p) = \sum_{i=1}^n u(x_i) g_i(p)$$

for some functions u and $g_i : \mathcal{L} \rightarrow [0, 1]$, $i = 1, \dots, n$, where $\sum_i g_i(p) = 1$. The g_i functions are meant to capture “distortions” in probabilities revealed by the person’s choices.³ (For example, the person’s choices might suggest that unlikely events are treated as even more unlikely than that “really” are.) The representation reduces to expected utility if $g_i(p) = p_i$ for $i = 1, \dots, n$. Since the g_i functions can be *non*-linear, the model departs from expected utility’s “affine in the probabilities” representation.

ADDENDUM II: MACHINA’S “PARADOX”

Here’s an example I learned from Mark Machina.⁴

Consider the following two distributions over some outcomes that matter to a person, Alice: I) a 9/10 chance of winning a trip to Venice, and a 1/10 chance of watching an excellent film about Venice (in a theater); and II) a 9/10 chance of winning a trip to Venice, and a 1/10 chance of spending the evening at home. Suppose that Alice strictly prefers the sure outcome “seeing an excellent film about Venice” to the sure outcome “staying at home.”

He argues that it is perfectly reasonable to Alice to prefer II) to I), seemingly contradicting the independence axiom; it is reasonable, he argues, because, if she does not win the trip to Venice—which had a 90 % chance—she would be too depressed to endure sitting through a film about Venice. It suggests that “what could have happened but didn’t” can reasonably influence preference over things that will happen. Indeed it does not seem outrageous that, if Alice were allowed to defer her choice about staying at home vs seeing the film until after the uncertainty about the trip is resolved, she would choose to stay at home—even though, absent uncertainty, she’d choose the film.

In my notation defining the independence axiom, r is “a trip to Venice for sure,” q is “watching an excellent film about Venice for sure” p is “staying at home for sure,” and $\alpha = .1$. The example questions whether the choice between p and q in isolation determines the conditional preference between p and q embedded in the bigger risk: in particular what the third distribution r is matters for the conditional preference between p and q in $\alpha [\cdot] + (1 - \alpha)r$.

³Specifically, if the outcomes are ranked from worst (x_1) to best (x_n), then RDEU sets $g_i(p) = h(\sum_{j=1}^i p_j) - h(\sum_{j=1}^{i-1} p_j)$, where $h : [0, 1] \rightarrow [0, 1]$ is increasing with $h(0) = 0$ and $h(1) = 1$. This structure is the source of the “cumulative” in cumulative prospect theory. Any “distortions” in the probabilities are captured by the single function h . If h is the identity function, then $g_i(p) = p_i$. In CPT, the g_i functions depend on whether outcome x_i is a gain or loss relative to some reference outcome; in particular there are two “ h ” functions, one for gains and another for losses: Wakker, Peter, and Amos Tversky. “An axiomatization of cumulative prospect theory.” *Journal of risk and uncertainty* 7.2 (1993): 147-175.

⁴It is a more-chaste variant of his example on page 1660 in Machina, Mark J. “Dynamic consistency and non-expected utility models of choice under uncertainty.” *Journal of Economic literature* 27.4 (1989): 1622-1668.